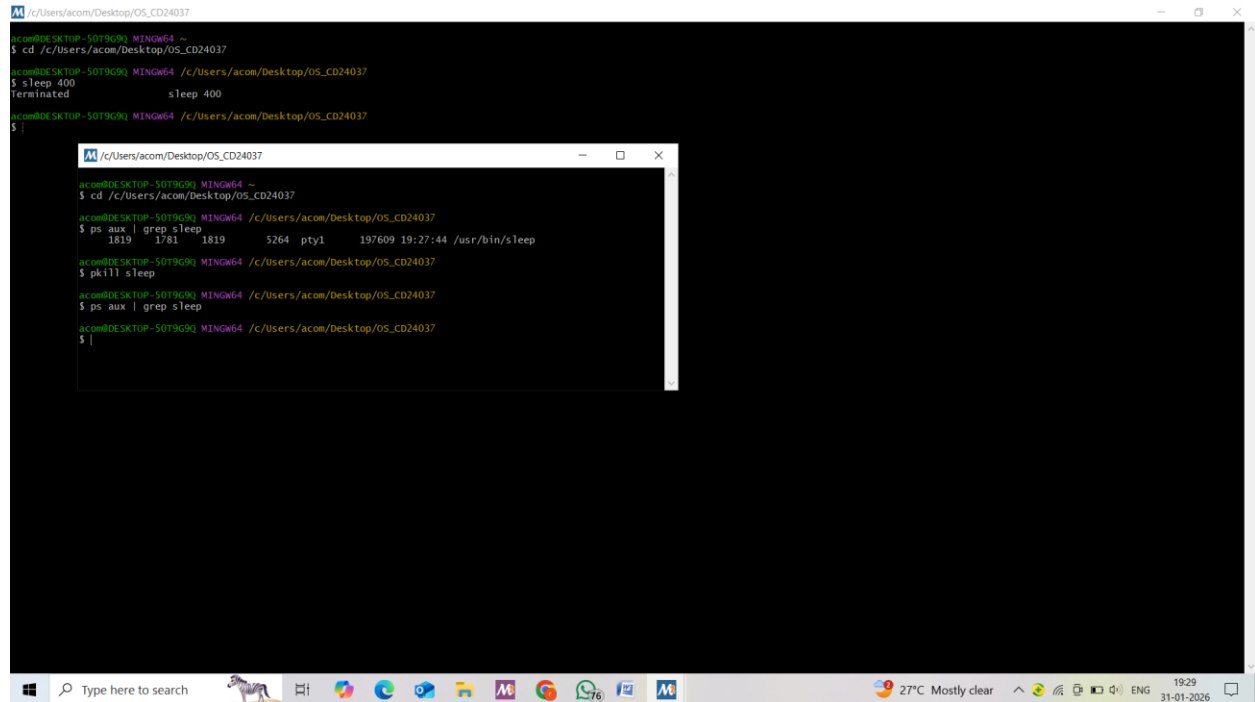


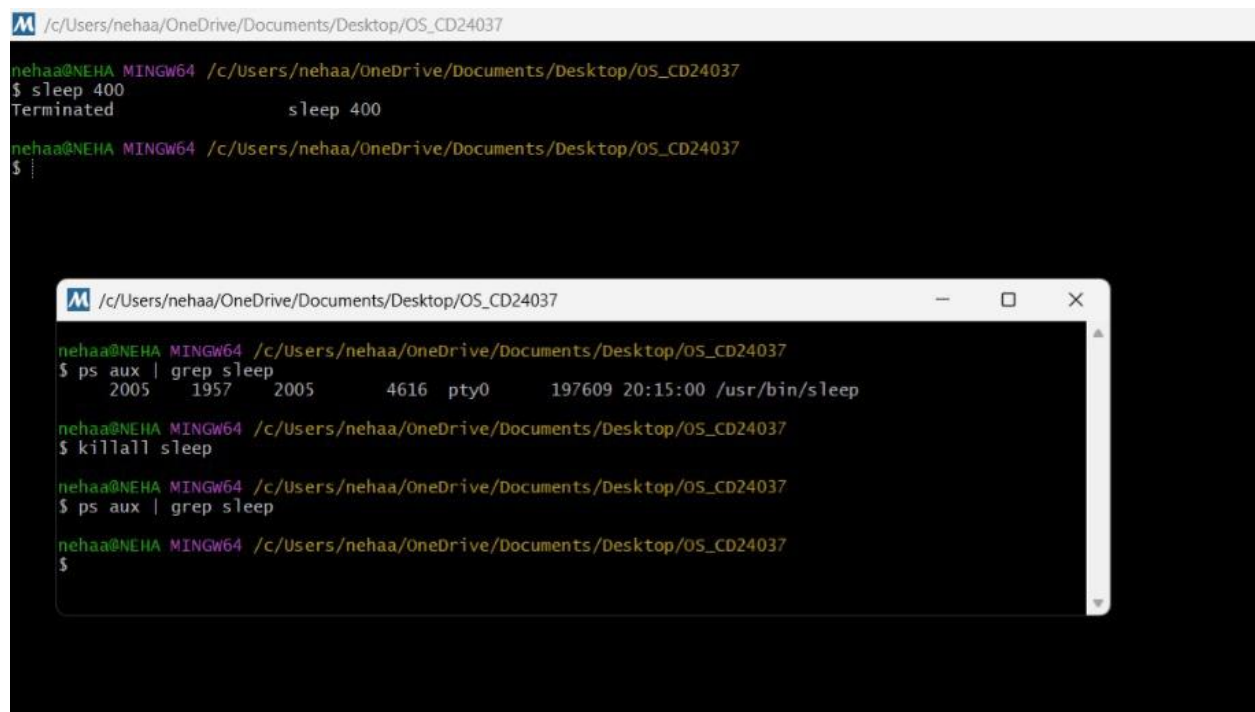
1. Adam is working in an IT company. He has been given a task to reduce the load of a system by killing some of the processes running in the LINUX operating system. Which commands will he use to complete the given task with the help of the following operation?

- Kill processes by name



```
acorn@DESKTOP-S0T9G9Q MINGW64 ~  
$ cd /c/Users/acorn/Desktop/OS_CD24037  
acorn@DESKTOP-S0T9G9Q MINGW64 /c/Users/acorn/Desktop/OS_CD24037  
$ sleep 400  
Terminated sleep 400  
acorn@DESKTOP-S0T9G9Q MINGW64 /c/Users/acorn/Desktop/OS_CD24037  
$  
acorn@DESKTOP-S0T9G9Q MINGW64 ~  
$ cd /c/Users/acorn/Desktop/OS_CD24037  
acorn@DESKTOP-S0T9G9Q MINGW64 /c/Users/acorn/Desktop/OS_CD24037  
$ ps aux | grep sleep  
1819 1781 1819 5264 ptY1 197609 19:27:44 /usr/bin/sleep  
acorn@DESKTOP-S0T9G9Q MINGW64 /c/Users/acorn/Desktop/OS_CD24037  
$ killall sleep  
acorn@DESKTOP-S0T9G9Q MINGW64 /c/Users/acorn/Desktop/OS_CD24037  
$ ps aux | grep sleep  
acorn@DESKTOP-S0T9G9Q MINGW64 /c/Users/acorn/Desktop/OS_CD24037  
$  
$
```

- Kill a process based on the process name



```
nehaa@NEHA MINGW64 /c/Users/nehaa/OneDrive/Documents/Desktop/OS_CD24037  
$ sleep 400  
Terminated sleep 400  
nehaa@NEHA MINGW64 /c/Users/nehaa/OneDrive/Documents/Desktop/OS_CD24037  
$  
nehaa@NEHA MINGW64 /c/Users/nehaa/OneDrive/Documents/Desktop/OS_CD24037  
$ ps aux | grep sleep  
2005 1957 2005 4616 pty0 197609 20:15:00 /usr/bin/sleep  
nehaa@NEHA MINGW64 /c/Users/nehaa/OneDrive/Documents/Desktop/OS_CD24037  
$ killall sleep  
nehaa@NEHA MINGW64 /c/Users/nehaa/OneDrive/Documents/Desktop/OS_CD24037  
$ ps aux | grep sleep  
nehaa@NEHA MINGW64 /c/Users/nehaa/OneDrive/Documents/Desktop/OS_CD24037  
$  
$
```

- Kill a single process at a time with the given process ID

```

/c/Users/acom/Desktop/OS_CD24037

acom@DESKTOP-50T9G9Q MINGW64 ~
$ cd /c/Users/acom/Desktop/OS_CD24037

acom@DESKTOP-50T9G9Q MINGW64 /c/Users/acom/Desktop/OS_CD24037
$ sleep 400
Terminated          sleep 400

acom@DESKTOP-50T9G9Q MINGW64 /c/Users/acom/Desktop/OS_CD24037
$ |

/c/Users/acom/Desktop/OS_CD24037

acom@DESKTOP-50T9G9Q MINGW64 ~
$ cd /c/Users/acom/Desktop/OS_CD24037

acom@DESKTOP-50T9G9Q MINGW64 /c/Users/acom/Desktop/OS_CD24037
$ ps aux | grep sleep
 682      644      682      10040  pty0      197609  20:21:20 /usr/bin/sleep

acom@DESKTOP-50T9G9Q MINGW64 /c/Users/acom/Desktop/OS_CD24037
$ kill 682

acom@DESKTOP-50T9G9Q MINGW64 /c/Users/acom/Desktop/OS_CD24037
$ ps aux | grep sleep

acom@DESKTOP-50T9G9Q MINGW64 /c/Users/acom/Desktop/OS_CD24037
$ |

```

2. Write a program for process creation using C

- Orphan Process

```

/c/Users/acom/Desktop/OS_CD24037

GNU nano 8.7
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        sleep(5);
        printf("Child Process\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());
    }
    else {
        printf("Parent exiting\n");
    }
    return 0;
}

```

```

/c/Users/acom/Desktop/OS_CD24037

acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ nano orphan.c

acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ gcc orphan.c -o orphan

acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ ./orphan
Parent exiting

acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ Child Process
PID : 1744
PPID : 1

```

- Zombie Process

 /c/Users/acom/Desktop/OS_CD24037

```
GNU nano 8.7
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        printf("Child exiting\n");
    }
    else {
        sleep(10);
        printf("Parent running\n");
    }
    return 0;
}
```

 /c/Users/acom/Desktop/OS_CD24037

```
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ nano zombie.c

acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ gcc zombie.c -o zombie

acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ gcc zombie.c -o zombie

acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ ./zombie
Child exiting
Parent running

acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ ps -el
  PID  PPID  PGID  WINPID  TTY      UID    STIME  COMMAND
  1524    1   1524    7940    ?        197609 18:23:51 /usr/bin/mintty
  1760   1525   1760    8064  pts/0    197609 18:39:01 /usr/bin/ps
  1525   1524   1525    7676  pts/0    197609 18:23:51 /usr/bin/bash

acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$
```

3. Create the process using fork () system call.

- Child Process creation
- Parent process creation
- PPID and PI

 /c/Users/acom/Desktop/OS_CD24037

```
GNU nano 8.7
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    pid = fork();

    if (pid < 0) {
        printf("Fork failed\n");
    }
    else if (pid == 0) {
        // Child process
        printf("Child Process\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());
    }
    else {
        // Parent process
        printf("Parent Process\n");
        printf("PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);
        wait(NULL);
    }
    return 0;
}
```

 /c/Users/acom/Desktop/OS_CD24037

```
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ nano fork.c

acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ gcc fork.c -o fork

acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ ./fork
Child Process
Parent Process
PID : 1770
PPID : 1769
PID : 1769
Child PID : 1770

acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037
$ |
```